



Day 8 Notes

Selfie Segmentation – Separating Me From the Background

★ Today = added the SECOND AI model. Now running Pose + Segmentation together, live, on the same webcam feed.

1 What is Selfie Segmentation?

A pretrained AI model that looks at each video frame and returns a MASK — basically a grayscale image where each pixel says "this is probably a person" or "this is probably background."

Different job than Pose: Pose finds JOINTS (points). Segmentation finds the whole SILHOUETTE (shape).

2 Loading the Segmentation Model

```
import { FilesetResolver, PoseLandmarker, ImageSegmenter } from
  "https://cdn.jsdelivr.net/npm/@mediapipe/tasks-vision@0.10.14"

imageSegmenter = await ImageSegmenter.createFromOptions(vision, {
  baseOptions: {
    modelAssetPath: '../selfie_segmenter.tflite'
  },
  runningMode: 'VIDEO',
  outputCategoryMask: true
})
```

Loaded alongside PoseLandmarker from Day 7 — both models run on the same video stream at the same time.

3 Running Segmentation Each Frame

```
const segResult = imageSegmenter.segmentForVideo(video, timestamp)
const mask = segResult.categoryMask.getAsFloat32Array()
```

- `segmentForVideo(...)` → runs segmentation on the current frame
- `categoryMask` → the result, one "category" value per pixel
- `getAsFloat32Array()` → reads it as a flat array of numbers I can loop through

4 Drawing the Mask (So I Can SEE It Working)

```

const imageData = ctx.createImageData(canvas.width, canvas.height)

for (let i = 0; i < mask.length; i++) {
  const isPerson = mask[i] === 0 // category 0 = person
  if (isPerson) {
    imageData.data[i*4] = 0 // R
    imageData.data[i*4+1] = 255 // G
    imageData.data[i*4+2] = 0 // B
    imageData.data[i*4+3] = 80 // Alpha (semi-transparent)
  }
}
ctx.putImageData(imageData, 0, 0)

```

This paints every "person" pixel green and semi-transparent, so I could visually confirm the mask lines up with my actual body on screen.

5 Combining Both Models

In the same detectLoop, I now:

- Run segmentation → paint green tint over my silhouette
- Run pose detection (from Day 7) → draw red dots on top, for joints

Result: green silhouette (WHERE I am) + red dots (WHERE my joints are), both updating live as I move.

Two AI models, two different kinds of info, same video frame.



Big takeaway:

Selfie Segmentation = per-pixel mask telling you person vs background. ImageData + putImageData = how to manually paint pixels onto a canvas based on data. Running two AI models together is totally normal — each gives different, complementary info. Next (Day 9): the actual MATH — affine transforms — to stretch a real clothing image onto my shoulder/hip points instead of just drawing dots.

— made while building: *AI Virtual Try-On E-commerce App* —